```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
df=pd.read_csv('/content/StudentPerformance.csv')
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 29 entries, 0 to 28
Data columns (total 6 columns):
 #   Column                 Non-Null Count  Dtype  
---  ------                 --------------  -----  
 0   Maths_Score            28 non-null     float64
 1   Reading_Score          27 non-null     float64
 2   Writing_Score          28 non-null     float64
 3   Placement_Score        27 non-null     float64
 4   Club_Join_Date         29 non-null     int64  
 5   Placement offer count  29 non-null     int64  
dtypes: float64(4), int64(2)
memory usage: 1.5 KB
```

```
df.isnull().sum()
```

	0
Maths_Score	1
Reading_Score	2
Writing_Score	1
Placement_Score	2
Club_Join_Date	0
Placement offer count	0

dtype: int64

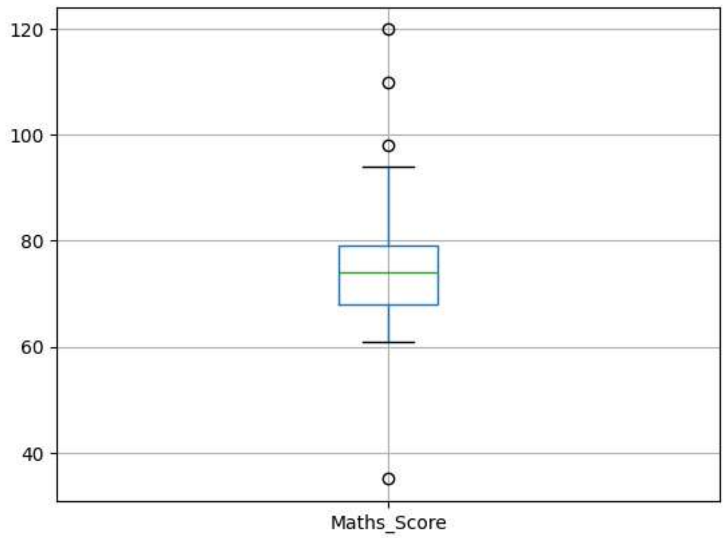
```
mean_val=df['Maths_Score'].mean()
df['Maths_Score']=df['Maths_Score'].fillna(mean_val)
mean_val=df['Reading_Score'].mean()
df['Reading_Score']=df['Reading_Score'].fillna(mean_val)
mean_val=df['Writing_Score'].mean()
df['Writing_Score']=df['Writing_Score'].fillna(mean_val)
mean_val=df['Placement_Score'].mean()
df['Placement_Score']=df['Placement_Score'].fillna(mean_val)
df.isnull().sum()
```

	0
Maths_Score	0
Reading_Score	0
Writing_Score	0
Placement_Score	0
Club_Join_Date	0
Placement offer count	0

dtype: int64

```
col=['Maths_Score']
df.boxplot(col)
```

```
<Axes: >
```




```
sorted_df=sorted(df['Maths_Score'])
sorted_df
q1=np.percentile(sorted_df,25)
q3=np.percentile(sorted_df,75)
print(q1,q3)
```

```
68.0 79.0
```

```
IQR=q3-q1
lwr_bound=q1-(1.5*IQR)
upr_bound=q3+(1.5*IQR)
print(lwr_bound,upr_bound)
```

```
51.5 95.5
```

```
sample_outliers=df[col][(df[col]<lwr_bound)|(df[col]>upr_bound)]
sample_outliers
```

	Maths_Score
0	NaN
1	NaN
2	NaN
3	35.0
4	NaN
5	NaN
6	NaN
7	NaN
8	NaN
9	NaN
10	NaN
11	NaN
12	NaN
13	NaN
14	NaN
15	110.0
16	NaN
17	NaN
18	NaN
19	120.0
20	NaN
21	NaN
22	NaN
23	NaN
24	NaN
25	NaN
26	NaN
27	98.0
28	NaN

```
median=np.median(df[col])
median
```

```
np.float64(74.0)
```

```
index_outliers = sample_outliers['Maths_Score'].dropna().index
for i in index_outliers:
  df.loc[i, col[0]] = median
df
```

	Maths_Score	Reading_Score	Writing_Score	Placement_Score	Club_Join_Date	Placement offer count
0	70.0	93.000000	61.000000	82.000000	2020	2
1	77.0	84.000000	65.000000	88.000000	2019	3
2	69.0	84.000000	68.000000	93.000000	2020	3
3	74.0	81.000000	73.000000	91.000000	2019	3
4	78.0	95.000000	73.000000	96.000000	2020	3
5	70.0	94.000000	69.000000	80.000000	2020	2